Financial Frictions, Market Power, and Innovation

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Motivation

- Innovation is a key driver of aggregate productivity growth
- But R&D is risky, intangible, and difficult to finance externally
- As a result, firms must rely heavily on internal funds to sustain innovation
- Market power can provide those funds by generating markup rents that are retained as equity
- How does market power affect innovation when firms face financial frictions?

Agg. Innovation Agg. Productivity Growth

This Paper

- Canonical theories mainly emphasize how market power affects **incentives** to innovate (Schumpeter, 1942; Arrow, 1962; Gilbert & Newbery, 1982; Romer, 1990; Aghion & Howitt, 1992; Aghion et al., 2005)
- I highlight a complementary channel: market power affects innovation through firms' internal **financing**
- Higher markups allow firms to accumulate internal funds, increasing their financial capacity to support innovation
- I document this mechanism using Portuguese firm-level data and quantify its aggregate importance in a macroeconomic model with endogenous innovation

Literature

Findings

In the Data

- Innovative firms are larger, more productive, and charge higher markups
- Innovative firms hold less debt and more equity
- Firms with more equity innovate more
- High-markup firms accumulate equity faster

In the Model

- Markup rents affect innovation through both incentives and financing
- Markup rents matter for aggregate outcomes even when the incentive channel is shut down
- Credit constraints amplify the importance of markup rents

Data

- Administrative firm-level data (CBHP) provided by Bank of Portugal
- Source is mandatory annual declaration (IES) to tax authority
- Covers the universe of non-financial firms in Portugal (2006–2023)
- Harmonized balance-sheet, income-statement, and employment information

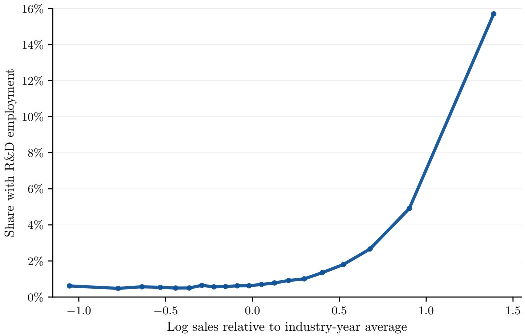
IES Form Sample Selection Data Cleaning Coverage Representativeness

Variables

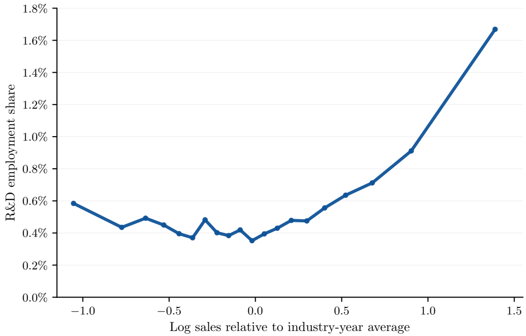
- **Production:** output, capital, labor, materials, employment
- **Innovation:** R&D employment
- **Markups and productivity:** firm-level estimates following the production-based approach (Hall, 1988; De Loecker & Warzynski, 2012; De Loecker et al., 2020)
- **Financial position:** debt, equity, cash, leverage, collateral
- **Firm controls:** age, exporter status, legal form, district, single/multiple owners, single/multi-establishments

Variable Definitions | Summary Stat. | Prod. Fun. Estimation | Prod. Fun. Estimates | Emp. Dist. | R&D Emp. Dist.

Fact 1: Innovation Is Concentrated in Large Firms



(a) Share with R&D employment



(b) R&D employment share

By Sector

Fact 2: Innovative Firms Are More Productive and Charge Higher Markups

$$\ln(Y_{it}) = \beta_0 + \beta_1 \ln(\text{R\&D emp}_{it}) + \Gamma' Z_{it} + \delta_s + \delta_t + \varepsilon_{it}$$

	Productivity	Markups
$\ln(\text{R\&D emp}_{it})$	0.0187**	0.0165**
	(0.0087)	(0.0081)

- For avg innovative firm: $9 \rightarrow 10$ R&D workers $\longleftrightarrow$ +0.20% productivity & +0.17% markups
- Results robust to controls for fixed costs
- Consistent with **incomplete pass-through**

Full Table Controlling for Overhead Passthrough

Fact 3: Innovative Firms Hold Less Debt and More Equity

	Leverage ratio	Equity ratio
$\mathbb{1}(\text{R\&D emp}_{it} > 0)$	-0.0143***	0.0135***
	(0.0036)	(0.0045)

- Innov firms have -1.4 pp (-6.5%) leverage & +1.4 pp (+3.2%) equity ratio
- Lower leverage not explained by cash holdings, collateral, or cost of debt
- Most of leverage differential from long-term debt
- Suggests innovation associated with **lower debt capacity**

Full Table Extensive Margin ST vs LT Leverage Cost of Debt

Fact 4: Firms with More Equity Innovate More

$$\text{Innovation}_{it} = \beta_0 + \beta_1 \ln(\text{Equity}_{it}) + \Gamma' Z_{it} + \delta_s + \delta_t + \varepsilon_{it}$$

	R&D participation	R&D employment share
$\ln(\text{Equity}_{it})$	0.0020***	0.0007***
	(0.0004)	(0.0002)

- +10% equity $\longleftrightarrow$ +0.02 pp (+1.1%) prob R&D & +0.007 pp (+1.3%) share R&D emp
- Robust to differences in leverage and collateral

Full Table Controlling for Leverage Controlling for Collateral

Fact 5: High Markup Firms Accumulate Equity Faster

$$\Delta \ln(\text{Equity}_{it}) = \beta_0 + \beta_1 \ln(\text{Markup}_{it}) + \Gamma' Z_{it} + \delta_s + \delta_t + \varepsilon_{it}$$

	All firms	Innovative firms
$\ln(\text{Markup}_{it})$	0.0955***	0.0757***
	(0.0124)	(0.0124)

- $+10\%$ markup $\longleftrightarrow$ $+0.9$ pp faster equity growth ($+0.7$ pp for innov firms)
- Not driven by lower investment, industry trends, or permanent productivity differences
- Consistent with **market power relaxing financing constraints**

Full Table Controlling for Investment Industry×Year FE Firm FE

Taking Stock

1. Innovation concentrated among large firms
2. Innovative firms more productive, higher markups, and incomplete pass-through
3. Innovative firms hold less debt and more equity
4. Firms with more equity innovate more
5. High-markup firms accumulate equity faster

Model Overview

Quantitative GE model with heterogeneous firms

Innovation

- Technology choice: traditional or R&D-intensive
- R&D raises productivity

Financial frictions

- Incomplete financial markets
- Borrowing limited by collateral

Market power

- Monopolistic competition
- Non-CES demand

Preferences Productivity

Technology

Traditional Technology

$$\max_{k,l,p} \pi^{\tau} = p(y^{\tau})y^{\tau} - wl - (r+\delta)k$$

$$y^{\tau} = A^{\tau}(z_p)k^{\alpha}l^{1-\alpha}$$

$$A^{\tau}(z_p) = \exp(z_p)$$

R&D-intensive Technology

$$\max_{k,l,v,p} \pi^{\kappa} = p(y^{\kappa})y^{\kappa} - wl - (1+\phi)wv - (r+\delta)k - \kappa$$

$$y^{\kappa} = A^{\kappa}(z_p, z_r, v)k^{\alpha}l^{1-\alpha}$$

$$A^{\kappa}(z_p, z_r, v) = \exp\left(z_p + z_r g(v)\right)$$

$$g(v) = \xi\frac{(1+v)^{1-\nu} - 1}{1-\nu}$$

FOCs

Technology

Traditional Technology

$$\max_{k,l,p} \pi^{\tau} = p(y^{\tau})y^{\tau} - wl - (r+\delta)k$$

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R&D-intensive Technology

$$\max_{k,l,v,p} \pi^{\kappa} = p(y^{\kappa})y^{\kappa} - wl - (1+\phi)w\boldsymbol{v} - (r+\delta)k - \kappa$$

R&D labor

$$y^{\kappa} = A^{\kappa}(z_p, z_r, v)k^{\alpha}l^{1-\alpha}$$

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R&D labor overhead

$$A^{\kappa}(z_p, z_r, v) = \exp\left(z_p + z_r g(v)\right)$$

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FOCs

Technology

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R&D-intensive Technology

$$\max_{k,l,v,p} \pi^{\kappa} = p(y^{\kappa})y^{\kappa} - wl - (1+\phi)wv - (r+\delta)k - \boldsymbol{\kappa}$$

R&D fixed cost

$$y^{\kappa} = A^{\kappa}(z_p, z_r, v)k^{\alpha}l^{1-\alpha}$$

$$A^{\kappa}(z_p, z_r, v) = \exp\left(z_p + z_r g(v)\right)$$

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FOCs

Technology

Traditional Technology

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$$g(v) = \xi\frac{(1+v)^{1-\nu} - 1}{1-\nu}$$

R&D ability

FOCs

Technology

Traditional Technology

$$\max_{k,l,p} \pi^{\tau} = p(y^{\tau})y^{\tau} - wl - (r+\delta)k$$

$$y^{\tau} = A^{\tau}(z_p)k^{\alpha}l^{1-\alpha}$$

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$$g(v) = \xi \frac{(1+v)^{1-\nu} - 1}{1-\nu}$$

R&D technology

FOCs

Financial Markets

- Firms can save and borrow using a one-period bond, with borrowing limited by pledgeable capital

- Letting equity be $a = k - b$, the constraint can be written as:

$$k \leq \lambda a$$

- Equity is valuable because it relaxes future borrowing constraints

Market Structure

- Firms are monopolistically competitive and produce differentiated varieties
- Final good producer aggregates differentiated varieties using Kimball demand
- Larger firms face lower demand elasticities and charge higher markups:

$$\mu(x) = \frac{\sigma(x)}{\sigma(x) - 1}, \qquad \sigma(x) = \theta x^{-\epsilon/\theta}, \qquad x \equiv \frac{y}{Y}$$

- Unlike CES demand, productivity improvements are only partially passed through into lower prices

Kimball Demand Variable Markups Pass-Through

Firm Problem

Let the individual state be $\mathbf{x} = (a, z_p, z_r)$. Firms choose between traditional and R&D-intensive technologies:

$$V(\mathbf{x}) = \max\left\{V^{\tau}(\mathbf{x}), V^{\kappa}(\mathbf{x})\right\}.$$

For each technology $j \in \{\tau, \kappa\}$:

$$V^{j}(\mathbf{x}) = \max_{c,a'}\left\{u(c) + \beta\mathbb{E}\left[V(\mathbf{x}') \mid \mathbf{x}\right]\right\}$$

subject to

$$c + a' = (1 + r)a + \pi^{j}(\mathbf{x}), \qquad c \geq 0, \qquad k \leq \lambda a.$$

- Static choices determine scale, markups, profits, and R&D intensity
- Dynamic choices determine how much earnings are retained inside the firm

Equilibrium

Mechanism

1. R&D raises productivity
2. More productive firms expand market shares
3. Under non-CES demand, larger market shares imply higher markups
4. Higher markups raise profits and retained earnings
5. More retained earnings relax future borrowing constraints
6. Relaxed financing constraints support further production and innovation

Quantitative Analysis

How important is the financing role of market power?

1. Calibrate the model to firm-level moments on innovation, financing, and markups
2. Validate the model using untargeted moments
3. Decompose the effects of markup rents into:
   - **incentives channel** → markup rents affect production and innovation decisions
   - **financing channel** → markup rents affect equity accumulation and borrowing capacity
4. Study how the financing role of market power varies with tightness of borrowing constraints

Model Fit

- The model captures well the concentration of economic activity, the aggregate importance of innovative firms, and the sorting of innovation toward large firms

Targeted moments (selec.)

Moment	Data	Model
Median markup	1.37	1.31
Leverage ratio	0.22	0.21
R&D elasticity wrt output	0.36	0.36
Std. dev. log productivity	0.94	0.87

Untargeted moments (selec.)

Moment	Data	Model
Top 20% output share	0.81	0.72
Innovative firms' emp. share	0.07	0.08
Innovating: top 5% firms	0.18	0.18
R&D intensity: top 5% firms	0.017	0.010

Fixed Parameters Fitted Parameters Validation

Decomposing the Financing Role of Market Power

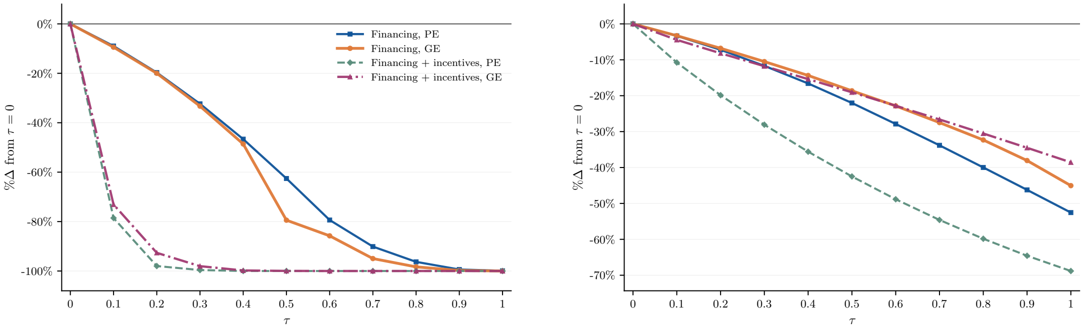
- Markup rents affect firms through two channels:
  - **Incentives:** higher rents increase the return to expanding and innovating
  - **Financing:** higher rents increase retained earnings and borrowing capacity
- I introduce an **internal-finance wedge** that limits how much markup rent can be retained:

$$\tilde{\pi}_{it}(\tau) = \pi_{it} - \tau \mathcal{M}_{it}, \qquad \tau \in [0, 1]$$

- I separate the two effects by varying which decisions internalize the wedge:
  - **Financing only:** operating policies fixed, dynamic saving re-optimized
  - **Financing + incentives:** operating and dynamic policies both re-optimized
- I conduct each exercise in PE and GE, yielding four counterfactuals.

Markup Rents Decomposition Design

Financing Role of Market Power



(a) R&D participation (b) Output

- Even when operating decisions are fixed, restricting retained markup rents reduces aggregate innovation and output.

R&D Labor

Financing Role of Market Power

- At $\tau = 0.1$, the financing channel accounts for roughly 11% of the decline in R&D participation and 31% of the decline in output in partial equilibrium.

	R&D participation	Output
PE	11%	31%
GE	13%	74%

- The importance of the financing channel rises as τ increases.

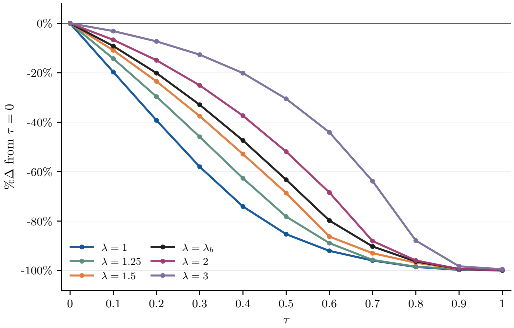
Full Table

Financing Role of Market Power

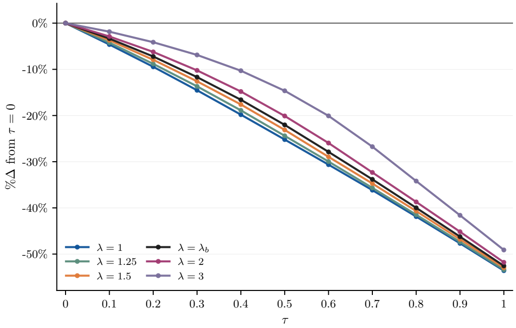
- General-equilibrium adjustment dampens incentive effects more than financing effects
  - Lower wages partially restore firms' incentives to produce and innovate
  - Higher interest rates raise returns on savings but do not restore lost equity
- The financing channel operates through firms' balance sheets
  - Restricting retained markup rents reduces aggregate equity
  - More firms become borrowing constrained
- Restricting retained markup rents reduces rents but average markups change little

Equilibrium Prices Equity & Constraints Rents vs Markups

Borrowing Constraints Amplify the Financing Channel



(a) R&D participation



(b) Output

- The aggregate cost of restricting markup rents is largest in financially constrained economies

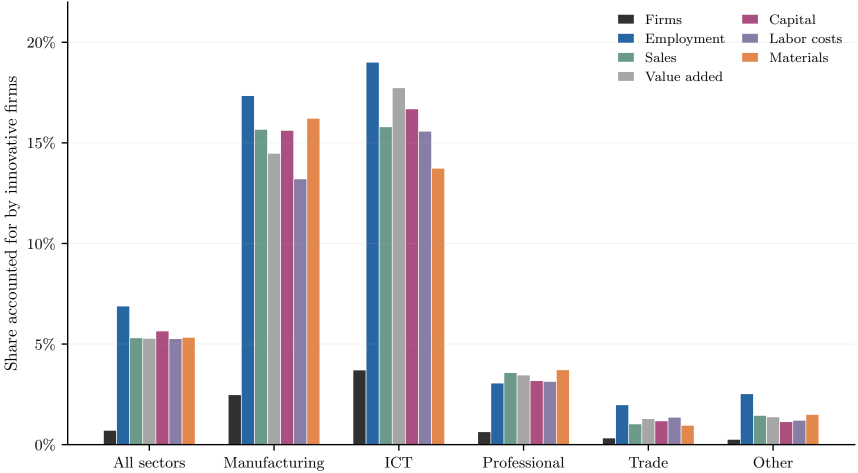
R&D Labor

Conclusion

- Markup rents finance future innovation by supporting equity accumulation
- A meaningful share of the effect of market power on innovation operates through the financing channel
- This channel becomes more important as borrowing constraints tighten
- Market power acts as an imperfect substitute for missing financial markets
- Reducing market power is less costly for innovation when firms have better access to external finance

Extra Slides

Aggregate Importance of Innovative Firms



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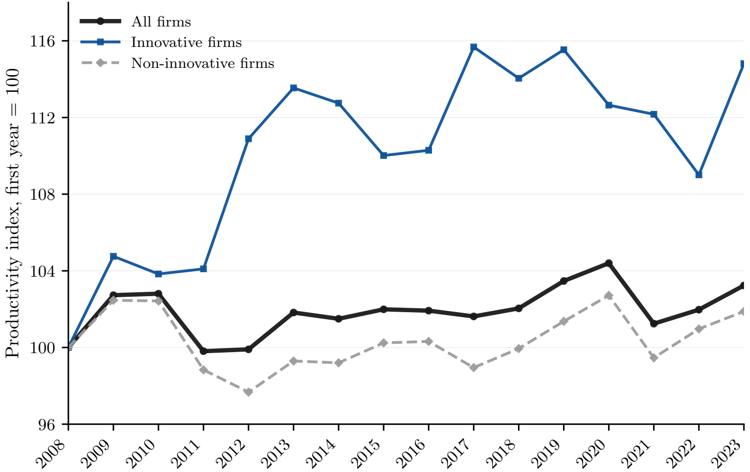
Aggregate Productivity Growth



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Contribution to the Literature

- **Competition and innovation:** Schumpeter (1942); Arrow (1962); Gilbert & Newbery (1982); Romer (1990); Aghion & Howitt (1992); Aghion et al. (2005)

  market power affects both innovation incentives and firms' financial capacity

- **Financial frictions and innovation:** Brown et al. (2009); Hall & Lerner (2010); Kerr & Nanda (2015)

  internal resources affect R&D investment when external finance is limited

- **Market power and agg. outcomes:** De Loecker et al. (2020); Autor et al. (2020); Edmond et al. (2023)

  markups affect productivity through a dynamic innovation-financing channel

- **Agg. effects of financial frictions:** Buera et al. (2011); Midrigan & Xu (2014); Moll (2014)

  financial frictions distort not only capital accumulation, but also R&D and productivity growth

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IES Form

IES

DECLARAÇÃO ANUAL

IES - INFORMAÇÃO EMPRESARIAL SIMPLIFICADA

(ENTIDADES RESIDENTES QUE EXERCEM, A TÍTULO PRINCIPAL, ACTIVIDADE COMERCIAL, INDUSTRIAL OU AGRÍCOLA E ENTIDADES NÃO RESIDENTES COM ESTABELECIMENTO ESTÁVEL)

IRC

ANEXO A

01 **N.º DE IDENTIFICAÇÃO FISCAL (NIPC)** 1

02 **EXERCÍCIO/PERÍODO** 1

02-A A contabilidade encontra-se organizada conforme:
(preencher apenas para os períodos de 2010 e seguintes)
1 NIC's (N) 2 NCRF's (S) 3 NCRF-PE (S) 4 NC-ME (M)

03 **DEMONSTRAÇÃO DOS RESULTADOS POR NATUREZAS - Exercícios de 2009 e anteriores**

		EXERCÍCIOS			
		N		N-1	
	CUSTOS E PERDAS	(1)	(2)	(3)	(4)
	Custo das mercadorias vendidas e das matérias consumidas:				
A0101	Mercadorias	. . ,		. . ,	
A0102	Matérias	. . ,	. . ,	. . ,	. . ,
A0103	Fornecimentos e serviços externos		. . ,		. . ,
	Custos com o pessoal:				
A0104	Remunerações	. . ,		. . ,	
	Encargos Sociais:				
A0105	Pensões	. . ,		. . ,	
A0106	Outros	. . ,	. . ,	. . ,	. . ,
A0107	Amortizações do imobilizado corpóreo e incorpóreo	. . ,		. . ,	
A0108	Ajustamentos	. . ,		. . ,	
A0109	Provisões	. . ,	. . ,	. . ,	. . ,

IES Form

0507 **NOTA 7 - PESSOAS AO SERVIÇO E HORAS TRABALHADAS**

RUBRICAS	NÚMERO MÉDIO DE PESSOAS	NÚMERO DE HORAS TRABALHADAS NO ANO
Pessoas ao serviço da empresa, remuneradas e não remuneradas:	A0417	A0426
Pessoas remuneradas ao serviço da empresa (assalariados):	A0419	A0428
Das quais: Aprendizes	A0420	
Trabalhadores no domicílio	A0421	
Pessoas não remuneradas ao serviço da empresa (não assalariados):	A1508	A1511
Pessoas ao serviço da empresa por tipo de horário:		
Pessoas ao serviço da empresa a tempo completo	A0418	A0427
Das quais: Pessoas remuneradas ao serviço da empresa a tempo completo	A1509	A1512
Pessoas ao serviço da empresa a tempo parcial	A1510	A1513
Das quais: Pessoas remuneradas ao serviço da empresa a tempo parcial	A0422	A0429
Pessoas ao serviço da empresa, das quais:		
Pessoas ao serviço da empresa afectas à investigação e desenvolvimento	A0424	
Prestadores de serviços	A0423	A0430
Pessoas colocadas através de agências de trabalho temporário	A0425	

Comentário:

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Variable Definitions

Main variables

Variable	Definition
Output	Sales of goods and services
Capital	Fixed assets
Labor	Employee expenses
Materials	COGS + supplies & ext. serv.
Employment	Number of employees
R&D emp.	Employees allocated to R&D
Debt	Total financial debt
Equity	Shareholder equity

Ratios and controls

Variable	Definition
Leverage	Debt / assets
Equity ratio	Equity / assets
Collateral	Tangible fixed assets / assets
Cash ratio	Cash & equivalents / assets
Overhead	Supp. & ext. serv. / expenses
Cost of debt	Interest expenses / debt
Long-term debt	Non-cur. fin. debt / debt
Age	Ref. year - Inc. year + 1
Exporter	Positive foreign sales or serv.

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Sample Selection

- Keep private and public limited-liability companies
- Restrict to domestically owned firms in the private sector
- Exclude:
  - Foreign branches
  - Holding companies
  - Public-sector entities
  - Financial firms
  - Inactive firms
- Retain standard annual declarations covering at least 360 days
- Restrict to mainland Portugal
- Require main industry to account for at least 75% of turnover

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Data Cleaning

- Drop observations with non-positive:
  - Sales
  - Materials
  - Capital
  - Labor
  - Equity
- Remove accounting inconsistencies:
  - Debt $>$ liabilities
  - Cash or fixed assets $>$ total assets
  - Labor or materials $>$ sales
  - Interest expenses $>$ debt
- Exclude extreme observations using within-industry screens based on:
  - Capital-to-sales
  - Labor-to-sales
  - Materials-to-sales
  - Debt-to-assets
  - Cash-to-assets
  - Fixed-assets-to-assets
- Winsorize all continuous variables at the 1st and 99th percentiles

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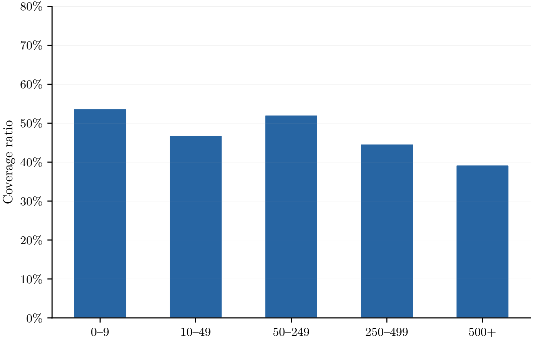
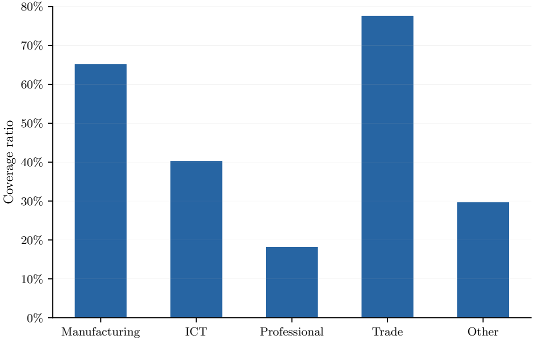
Coverage



By firm size



By sector

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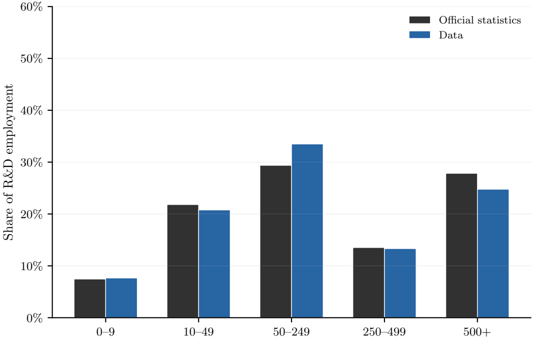
Representativeness



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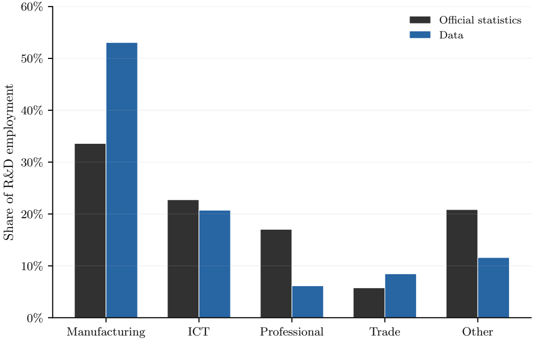

By sector

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Summary Statistics

	Mean	SD	p10	p25	p50	p75	p90
Output (000)	1,121	3,009	53	108	260	739	2,233
Capital (000)	329	1,019	2	10	42	178	644
Labor (000)	193	450	12	25	59	151	399
Materials (000)	752	2,035	21	53	153	484	1,574
Debt (000)	296	952	0	2	35	152	561
Equity (000)	467	1,352	11	30	91	283	906
Age	15.92	12.99	3	7	13	22	32
Exporter	0.23	0.42	0	0	0	0	1
Single-owner	0.23	0.42	0	0	0	0	1
Multi-establishment	0.13	0.34	0	0	0	0	1
Leverage ratio	0.22	0.22	0.00	0.00	0.16	0.35	0.54
Equity ratio	0.42	0.27	0.09	0.20	0.39	0.63	0.83
Collateral ratio	0.26	0.24	0.01	0.06	0.18	0.40	0.64
Cash ratio	0.20	0.22	0.01	0.03	0.11	0.30	0.55
Overhead ratio	0.28	0.20	0.07	0.13	0.22	0.40	0.59

Notes: All financial variables are deflated to real 2023 euros. $N = 2{,}631{,}219$ firm-year observations.

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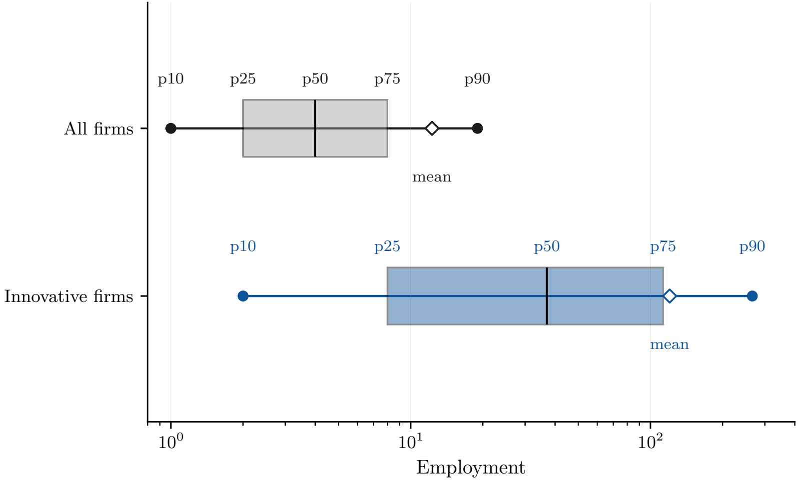
Employment Distribution



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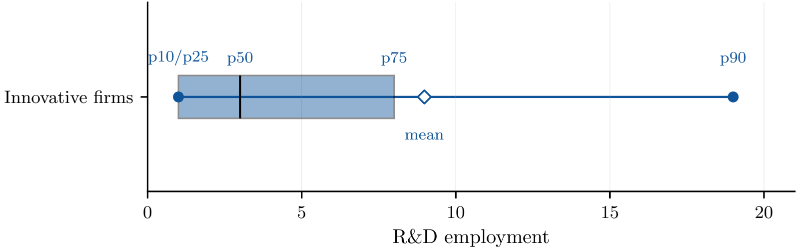
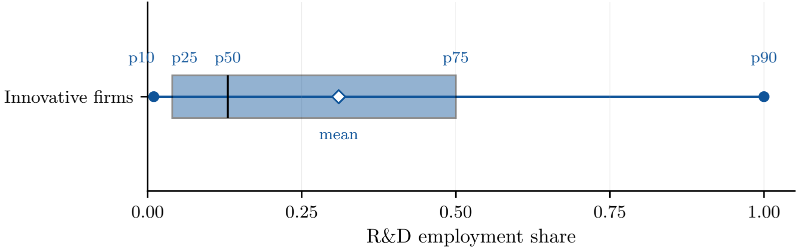
R&D Employment Distribution





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Production Function Estimation

- Production-based approach to estimate firm-level productivity and markups (Hall, 1988; De Loecker & Warzynski, 2012)

- Firm production with materials as the flexible input and capital predetermined

$$y_{it} = f(\ell_{it}, k_{it}) + \omega_{it} + \varepsilon_{it}$$

- Production functions estimated separately by 2-digit industry using control-function methods (Levinsohn & Petrin, 2003; Ackerberg et al., 2015)

- Firm-level productivity:

$$\widehat{\omega}_{it} = y_{it} - \widehat{f}(\ell_{it}, k_{it})$$

- Firm-level markups recovered from static cost minimization:

$$\mu_{it} = \frac{\theta_{m,it}}{\alpha_{m,it}}$$

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Production Function Estimates

	Mean	SD	p10	p25	p50	p75	p90
Returns to scale	0.92	0.08	0.82	0.89	0.93	0.95	1.02
Markups	1.73	1.07	1.01	1.13	1.37	1.86	2.84
Productivity	0.95	0.53	0.28	0.74	0.91	1.10	1.39

Notes: Markups are computed as $\mu_{it} = \theta_{m,it}/\alpha_{m,it}$. Productivity is reported in levels as $\exp(\widehat{\omega}_{it})$. $N = 1{,}550{,}506$ firm-year observations.

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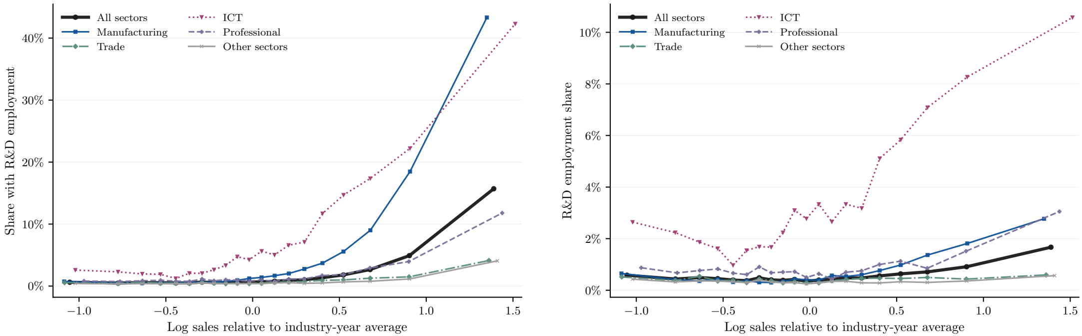
R&D Employment and Firm Size by Sector



(a) Share with R&D employment

(b) R&D employment share

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Innovation and Firm Outcomes: Full Table

	$\ln(\text{Productivity}_{it})$		$\ln(\text{Markup}_{it})$	
	(1)	(2)	(3)	(4)
$\ln(\text{R\&D emp}_{it})$	0.0199**	0.0187**	0.0180*	0.0165**
	(0.0096)	(0.0087)	(0.0094)	(0.0081)
Industry FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Firm controls	Y	Y	Y	Y
Overhead ratio	-	Y	-	Y
Observations	13,478	13,478	13,478	13,478
R-squared	0.8243	0.8358	0.3842	0.4288

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Incomplete Pass-Through of Innovation Gains

$$p = \mu \times MC \qquad\qquad d \ln p = d \ln \mu + d \ln MC$$

Under Cobb–Douglas production: $d \ln MC = -d \ln z$

	Productivity	Markups
Effect of +1% R&D employment	+0.0187%	+0.0165%

$$d \ln p = 0.0165 - 0.0187 = -0.0022$$

- Innovation ↓ marginal costs but also ↑ markups
- ≈ 88% of innovation gains absorbed into higher markups

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Innovation and Financing: Full Table

	Leverage ratio$_{it}$			Equity ratio$_{it}$		
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}(\text{R\&D emp}_{it} > 0)$	-0.0143***	-0.0122***	-0.0084**	0.0135***	0.0100**	0.0084*
	(0.0036)	(0.0035)	(0.0033)	(0.0045)	(0.0047)	(0.0047)
Industry FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Firm controls	Y	Y	Y	Y	Y	Y
Cash ratio	-	Y	Y	-	Y	Y
Collateral ratio	-	-	Y	-	-	Y
Observations	1,074,551	1,074,551	1,074,551	1,074,551	1,074,551	1,074,551
R-squared	0.0587	0.1124	0.1505	0.1191	0.2136	0.2184

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Innovation and Financing: Extensive Margin

	$\mathbb{1}(\text{Total debt}_{it} > 0)$			$\mathbb{1}(\text{Long-term debt}_{it} > 0)$		
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}(\text{R\&D emp}_{it} > 0)$	-0.0467***	-0.0431***	-0.0401***	-0.0166**	-0.0134*	-0.0087
	(0.0103)	(0.0090)	(0.0094)	(0.0076)	(0.0077)	(0.0076)
Industry FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Firm controls	Y	Y	Y	Y	Y	Y
Cash ratio	-	Y	Y	-	Y	Y
Collateral ratio	-	-	Y	-	-	Y
Observations	1,074,551	1,074,551	1,074,551	1,074,551	1,074,551	1,074,551
R-squared	0.1378	0.1789	0.1853	0.1262	0.1493	0.1603

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Innovation and Financing: Short vs Long Term

	Short-term leverage ratio$_{it}$		Long-term leverage ratio$_{it}$	
	(1)	(2)	(3)	(4)
$\mathbb{1}(\text{R\&D emp}_{it}) > 0)$	-0.0054***	-0.0044***	-0.0130***	-0.0080***
	(0.0015)	(0.0016)	(0.0031)	(0.0028)
Industry FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Firm controls	Y	Y	Y	Y
Cash ratio	-	Y	-	Y
Collateral ratio	-	Y	-	Y
Observations	1,074,551	1,074,551	1,074,551	1,074,551
R-squared	0.1232	0.1370	0.0459	0.1225

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Innovation and Financing: Cost of Debt

	Cost of debt_{it}		
	(1)	(2)	(3)
$\mathbb{1}(\text{R\&D emp}_{it} > 0)$	-0.0027*	-0.0023*	-0.0023*
	(0.0015)	(0.0014)	(0.0014)
Industry FE	Y	Y	Y
Year FE	Y	Y	Y
Firm controls	Y	Y	Y
Collateral ratio	-	Y	Y
Leverage ratio	-	Y	Y
Maturity structure	-	-	Y
Observations	831,453	831,453	831,453
R-squared	0.0634	0.1145	0.1161

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Innovation and Equity: Full Table

	$\mathbb{1}(\text{R\&D emp}_{it} > 0)$		$\frac{\text{R\&D emp}_{it}}{\text{Employment}_{it}}$	
	(1)	(2)	(3)	(4)
$\ln(\text{Equity}_{it})$	0.0020***	0.0019***	0.0007***	0.0006***
	(0.0004)	(0.0005)	(0.0002)	(0.0002)
Industry FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Firm controls	Y	Y	Y	Y
Collateral ratio	-	Y	-	Y
Leverage ratio	-	Y	-	Y
Observations	1,074,551	1,074,551	1,062,142	1,062,142
R-squared	0.1228	0.1229	0.0208	0.0209

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Markups and Equity: Full Table

	$\Delta \ln(\text{Equity}_{it})$			
	All firms		*Innovative firms*	
	(1)	(2)	(3)	(4)
$\ln(\text{Markup}_{it})$	0.0955***	0.0920***	0.0757***	0.0682***
	(0.0124)	(0.0123)	(0.0124)	(0.0119)
Industry FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Firm controls	Y	Y	Y	Y
Collateral ratio	-	Y	-	Y
Leverage ratio	-	Y	-	Y
Observations	1,550,506	1,550,506	13,478	13,478
R-squared	0.0341	0.0451	0.0399	0.0472

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Markups and Equity: Controlling for Investment

	Δ ln(Equity$_{it}$)			
	All firms		*Innovative firms*	
	(1)	(2)	(3)	(4)
ln(Markup$_{it}$)	0.0920***	0.0947***	0.0757***	0.0759***
	(0.0123)	(0.0123)	(0.0124)	(0.0115)
Industry FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Firm controls	Y	Y	Y	Y
Collateral ratio	Y	Y	Y	Y
Leverage ratio	Y	Y	Y	Y
Investment rate	-	Y	-	Y
Observations	1,550,506	1,550,506	13,478	13,478
R-squared	0.0451	0.0531	0.0472	0.0721

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Markups and Equity: Additional Fixed Effects

	$\Delta \ln(\text{Equity}_{it})$					
	All firms			*Innovative firms*		
	(1)	(2)	(3)	(4)	(5)	(6)
$\ln(\text{Markup}_{it})$	0.0920***	0.0917***	0.3018***	0.0682***	0.0673***	0.1847***
	(0.0123)	(0.0115)	(0.0357)	(0.0119)	(0.0116)	(0.0373)
Industry FE	Y	-	-	Y	-	-
Year FE	Y	-	-	Y	-	-
Industry $\times$ Year FE	-	Y	Y	-	Y	Y
Firm FE	-	-	Y	-	-	Y
Firm controls	Y	Y	Y	Y	Y	Y
Collateral ratio	Y	Y	Y	Y	Y	Y
Leverage ratio	Y	Y	Y	Y	Y	Y
Observations	1,550,506	1,550,500	1,504,447	13,478	13,365	11,639
R-squared	0.0451	0.0520	0.2694	0.0472	0.1102	0.3760

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Preferences

Firm owners maximize expected lifetime utility:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it})$$

Period utility is CRRA:

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}$$

- β: discount factor
- γ: coefficient of relative risk aversion

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Productivity

Firms are heterogeneous in productivity and R&D ability:

$$z_{p,it} = \rho z_{p,it-1} + \sigma \varepsilon_{it}, \qquad \varepsilon_{it} \sim N(0,1)$$

$$z_{r,i} \sim \text{Pareto}(b_r)$$

- z_p: persistent productivity
- z_r: permanent R&D ability

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Kimball Demand

Final-good aggregation:

$$\int \Upsilon\left(\frac{y_i}{Y}\right) di = 1$$

Following Klenow & Willis (2016), the Kimball aggregator is:

$$\Upsilon(x) = 1 + (\theta - 1)e^{1/\epsilon}\epsilon^{\frac{\theta}{\epsilon}-1}\left[\Gamma\left(\frac{\theta}{\epsilon}, \frac{1}{\epsilon}\right) - \Gamma\left(\frac{\theta}{\epsilon}, \frac{x^{\epsilon/\theta}}{\epsilon}\right)\right].$$

Demand elasticity:

$$\sigma(x) = \theta x^{-\epsilon/\theta}, \qquad x = \frac{y_i}{Y}.$$

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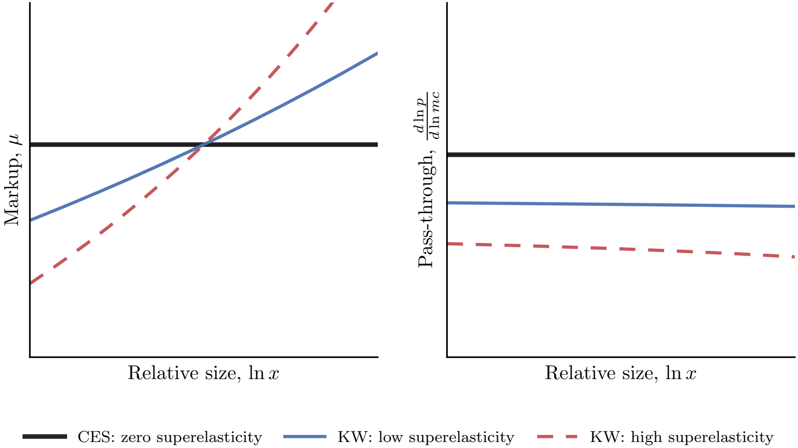
Variable Markups



Incomplete Pass-Through in the Model

Pricing satisfies

$$p(x) = \mu(x)mc.$$

With Kimball demand, markups vary with firm size, so marginal-cost changes are not fully passed through into prices:

$$\rho(x) \equiv \frac{d \ln p(x)}{d \ln mc} = \frac{1}{1 + \frac{\epsilon}{\theta} \frac{\sigma(x)}{\sigma(x)-1}}.$$

- Under CES demand, $\epsilon = 0$, so $\rho(x) = 1$
- Under Kimball demand, $\epsilon > 0$, so $\rho(x) < 1$
- Productivity gains partly translate into higher markups rather than lower prices

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Static Optimality Conditions

Let $q = y/Y$. Marginal revenue is

$$MR(q) = p(q)\left(1 - \frac{1}{\sigma(q)}\right) = \frac{p(q)}{\mu(q)}.$$

Traditional technology

$$MR(q)MPK^{\tau} = r + \delta, \qquad MR(q)MPL^{\tau} = w.$$

R&D-intensive technology

$$MR(q)MPK^{\kappa} = r + \delta, \qquad MR(q)MPL^{\kappa} = w,$$

$$MR(q)MPV^{\kappa} = (1+\phi)w, \qquad MPV^{\kappa} = z_r g'(v) y^{\kappa}.$$

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Stationary Equilibrium

A stationary equilibrium consists of aggregate variables $\mathbf{\Omega} = (w, r, P, Y, D)$, policy functions $\mathcal{T}(\mathbf{x}), k(\mathbf{x}), l(\mathbf{x}), v(\mathbf{x}), p(\mathbf{x}), c(\mathbf{x}), a'(\mathbf{x})$, value functions, and invariant distribution $\Lambda(\mathbf{x})$, such that:

1. Firms solve their dynamic optimization problem
2. Labor market clears:
$$\int [l(\mathbf{x}) + v(\mathbf{x})] d\Lambda(\mathbf{x}) = \bar{L}$$
3. Capital market clears:
$$\int a \, d\Lambda(\mathbf{x}) = \int k(\mathbf{x}) d\Lambda(\mathbf{x})$$
4. Kimball aggregation conditions hold
5. $\Lambda(\mathbf{x})$ is invariant

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Externally Fixed Parameters

Parameter	Value	Description
α	0.33	Capital share
γ	1.50	Risk aversion
δ	0.06	Depreciation
β	0.865	Discount factor
$\bar{L}$	1.00	Labor supply

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Internally Fitted Parameters

Parameter	Description	Targeted moment	Model	Data
Innovation				
ξ	R&D technology scale	Relative employment: innov./all	9.146	9.790
ν	R&D technology curvature	Elasticity $\ln(RD)$ on $\ln(Y)$	0.362	0.359
κ	Innovation fixed cost	Share innovating firms	0.009	0.020
ϕ	R&D labor overhead	R&D employment share	0.003	0.005
b_r	R&D ability dispersion	Top 1% R&D employment share	0.999	0.910
Productivity				
ρ	Productivity persistence	Top 10% employment share	0.474	0.660
σ	Productivity volatility	Std. dev. $\ln(Z)$	0.868	0.940
Market power				
θ	Baseline demand elasticity	Median markup	1.307	1.370
ϵ/θ	Demand superelasticity	Mean markup	1.315	1.730
Financing				
λ	Borrowing constraint	Mean leverage ratio	0.209	0.220

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Validation Moments

Moment	Data	Model
Concentration		
Top 20% output share	0.810	0.718
Top 10% output share	0.680	0.523
Top 5% output share	0.540	0.327
Aggregate importance of innovative firms		
Employment share	0.070	0.081
Output share	0.050	0.097
Innovation across the size distribution		
Share innovating: top 50%	0.030	0.018
Share innovating: top 20%	0.070	0.044
Share innovating: top 10%	0.110	0.088
Share innovating: top 5%	0.180	0.182
R&D intensity: top 50%	0.007	0.003
R&D intensity: top 20%	0.010	0.004
R&D intensity: top 10%	0.014	0.006
R&D intensity: top 5%	0.017	0.010

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Markup Rents

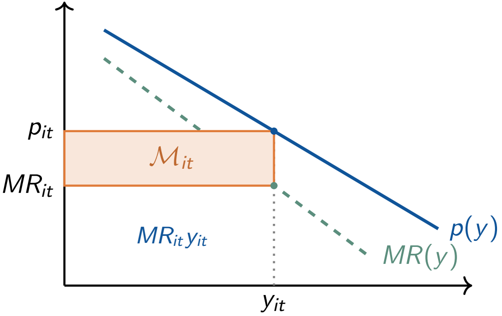
Markup rents are the portion of revenue generated by market power:

$$\mathcal{M}_{it} = \left(1 - \frac{1}{\mu_{it}}\right) p_{it} y_{it}$$

To isolate the financing role of markup rents, I introduce an **internal-finance wedge**:

$$\tilde{\pi}_{it} = \pi_{it} - \tau \mathcal{M}_{it}$$

- $\tau = 0$: baseline economy
- $\tau = 1$: markup rents cannot be retained as equity



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Counterfactual Design

Counterfactual	Operating policies	Dynamic policies	Aggregate state
Financing, PE	Baseline	Re-optimized	Fixed
Financing, GE	Baseline	Re-optimized	Adjusted
Financing + incentives, PE	Re-optimized	Re-optimized	Fixed
Financing + incentives, GE	Re-optimized	Re-optimized	Adjusted

- **Operating policies:** technology choice, inputs, prices, output, and R&D
- **Dynamic policies:** consumption and equity accumulation
- **PE vs GE:** aggregate prices fixed versus endogenous

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Quantitative Importance of the Financing Channel

	R&D participation		Output	
τ	PE	GE	PE	GE
0.1	0.11	0.13	0.31	0.74
0.2	0.20	0.22	0.36	0.83
0.3	0.33	0.34	0.42	0.89
0.4	0.47	0.49	0.47	0.93
0.5	0.63	0.80	0.52	0.98

Notes: Entries report the share of the total effect attributable to the financing channel. For each outcome, the share is computed as the ratio of the absolute percentage change in the financing-only exercise to the absolute percentage change in the corresponding financing-and-incentives exercise. PE denotes partial equilibrium and GE denotes general equilibrium.

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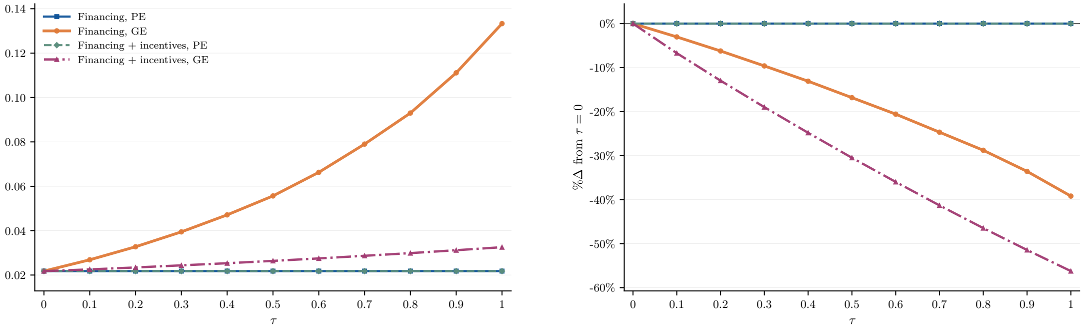
Equilibrium Prices



(a) Interest rate (b) Wage

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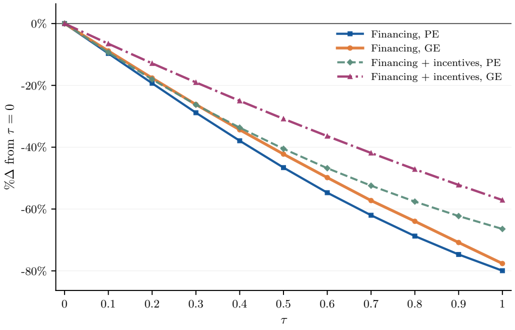
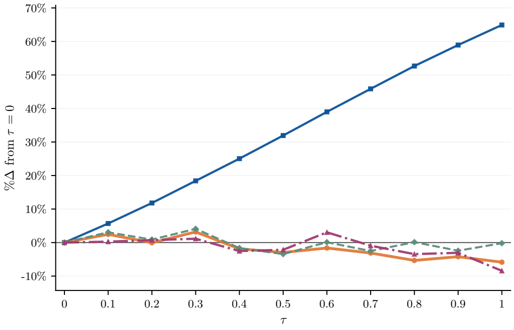
Equity and Financing Constraints



(a) Equity



(b) Share constrained

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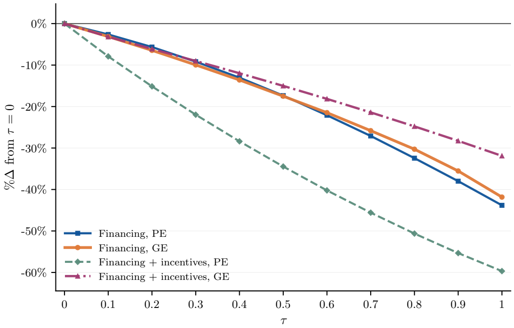
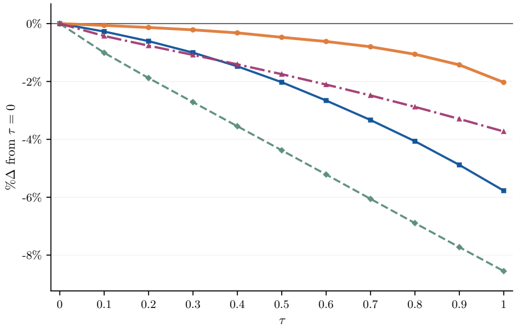
Markup Rents versus Average Markups



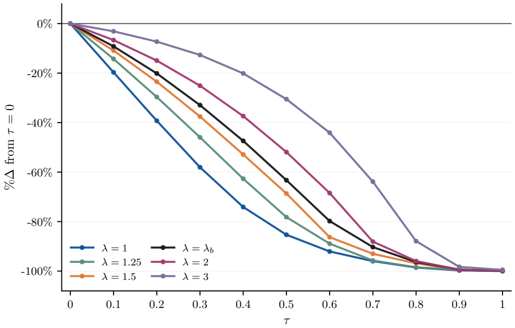
(a) Markup rents



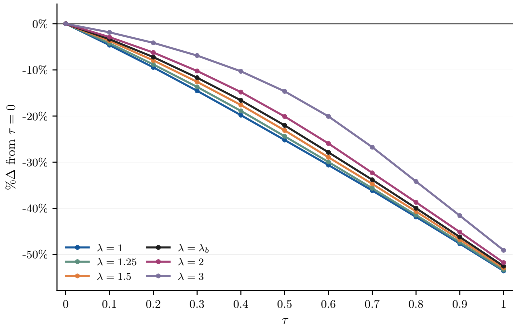
(b) Revenue-weighted markups

Unweighted Markups Back

Alternative Borrowing Constraints



(a) R&D participation



(b) Output

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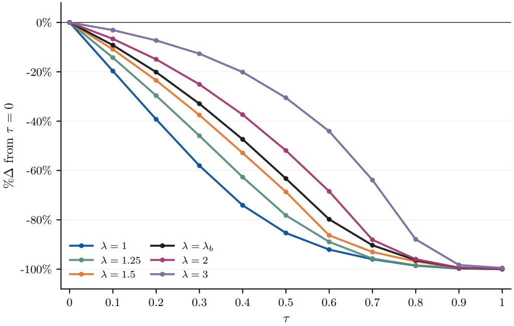
Aggregate R&D Labor under Alternative Borrowing Constraints



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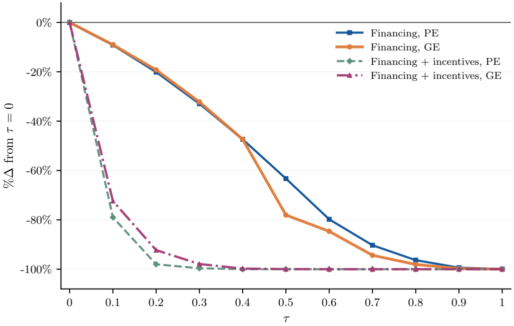
Aggregate R&D Labor



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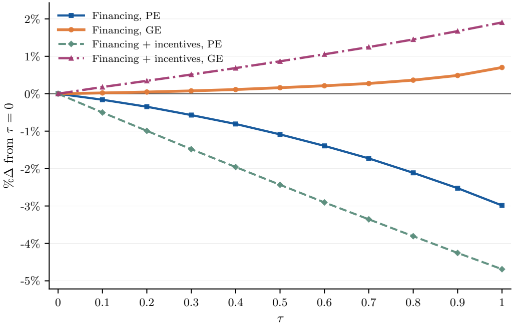
Unweighted Average Markups



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